

```
kalpesh@drparadox:~$ ifconfig
enp1s0  Link encap:Ethernet  HWaddr 20:1a:06:a5:b7:38
        inet addr:10.42.0.1  Bcast:10.42.0.255  Mask:255.255.255.0
        inet6 addr: fe80::add5:d5ae:b83a:e970/64  Scope:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:97 errors:0 dropped:0 overruns:0 frame:0
        TX packets:98 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:14541 (14.5 KB)  TX bytes:13600 (13.6 KB)
        Interrupt:33
```

Fig. 3: ifconfig command

```
kalpesh@drparadox:~$ nmap -n -sP 10.42.0.255/24

Starting Nmap 7.01 ( https://nmap.org ) at 2017-04-01 12:34 IST
Nmap scan report for 10.42.0.1
Host is up (0.00085s latency).
Nmap scan report for 10.42.0.193
Host is up (0.00094s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.06 seconds
```

Fig. 4: Searching for IP address of Raspberry Pi

```
pi@raspberrypi:~$ cat .vnc/xstartup
#!/bin/sh

xsetroot -solid grey
autocutsel -fork
#x-terminal-emulator -geometry 80x24+10+10 -ls -title 'VNCDESKTOP Desktop' &
#x-window-manager &
# Fix to make GNOME work
export XKL_XMOUSEMAP_DISABLE=1
/etc/X11/Xsession
pi@raspberrypi:~$
```

Fig. 5: Viewing contents of modified xstartup file

Bcast. This should be in the second line when ifconfig command is given.

Search for the IP address assigned to Raspberry Pi using `$ nmap -n -sP 10.42.0.255/24` command. Replace 10.42.0.255 with Bcast address noted in the previous step. nmap tool shows the details of all hosts that are up and running, along with their IP addresses. Make a note of the IP address (here, it is 10.42.0.193) assigned to Raspberry Pi.

Login to Raspberry Pi using `$ ssh -Y pi@10.42.0.193` command. Replace 10.42.0.193 with the IP address noted using nmap tool. Enter default password as raspberry.

This setup allows you to access Raspberry Pi anytime over SSH using

the IP address assigned to it, in non-graphical command-line mode. Note that, accessing the graphical desktop requires installing additional packages such as VNC server. This is explained next.

Installing required packages on Raspberry Pi

Internet access is available on Raspberry Pi since Ubuntu connected to a Wi-Fi network has shared its connection over Ethernet. Run the following command to update the packages list from the repositories:

```
$ sudo apt-get update
```

Graphical desktop of Raspberry Pi can be shared using tightvncserver package. For that, run the following command:

```
$ sudo apt-get install tightvncserver
```

Run the following command to perform the initial set up of tightvncserver:

```
$ tightvncserver
```

Enter a suitable password for future connections. Enter n for No to view only password.

Run the following command to copy and paste from VNC server:

```
$ sudo apt-get install autocutsel
```

Add autocutsel -fork in `/home/pi/.vnc/xstartup` using the following command:

```
$ sudo nano /home/pi/.vnc/xstartup
```

Save it using Ctrl + x.

Contents of the modified xstartup file can be viewed using the following command:

```
$ cat .vnc/xstartup
```

Restart VNC server for autocutsel to take effect, using the following command:

```
$ vncserver -kill :1 then command $
vncserver :1
```

Accessing Raspberry Pi using VNC Viewer

Although Raspberry Pi has the required packages installed to stream its graphical desktop, VNC Viewer is needed to access it. Since we are using an Ubuntu system to access Raspberry Pi, install VNC Viewer using the following command:

```
$ sudo apt-get install ssvnc
```

You may need to install xtightvncviewer using the following command:

```
$ sudo apt-get install xtightvncviewer
```

It is now possible to access the graphical desktop of Raspberry Pi using the following command:

```
$ xtightvncviewer 192.168.0.193:1
```

Replace 192.168.0.193 with the IP address of your Raspberry Pi. Enter the password to access it. Alternatively, the same results can be obtained using RealVNC Viewer package available on www.realvnc.com

Raspberry Pi can be logged on over SSH using an Ethernet network with Ubuntu system. Graphical desktop of Raspberry Pi running VNC server can be accessed using VNC Viewer. Internet can be accessed on Raspberry Pi by sharing the Wi-Fi connection of Ubuntu system. This configuration lets you access Raspberry Pi from any remote location using a laptop and a regular Ethernet cable.

Note that, vncserver does not start automatically when Raspberry Pi boots up. You need to start it using the command given below. Use SSH every time you want to access the graphical desktop of Raspberry Pi.

```
$ vncserver :1 EFY
```

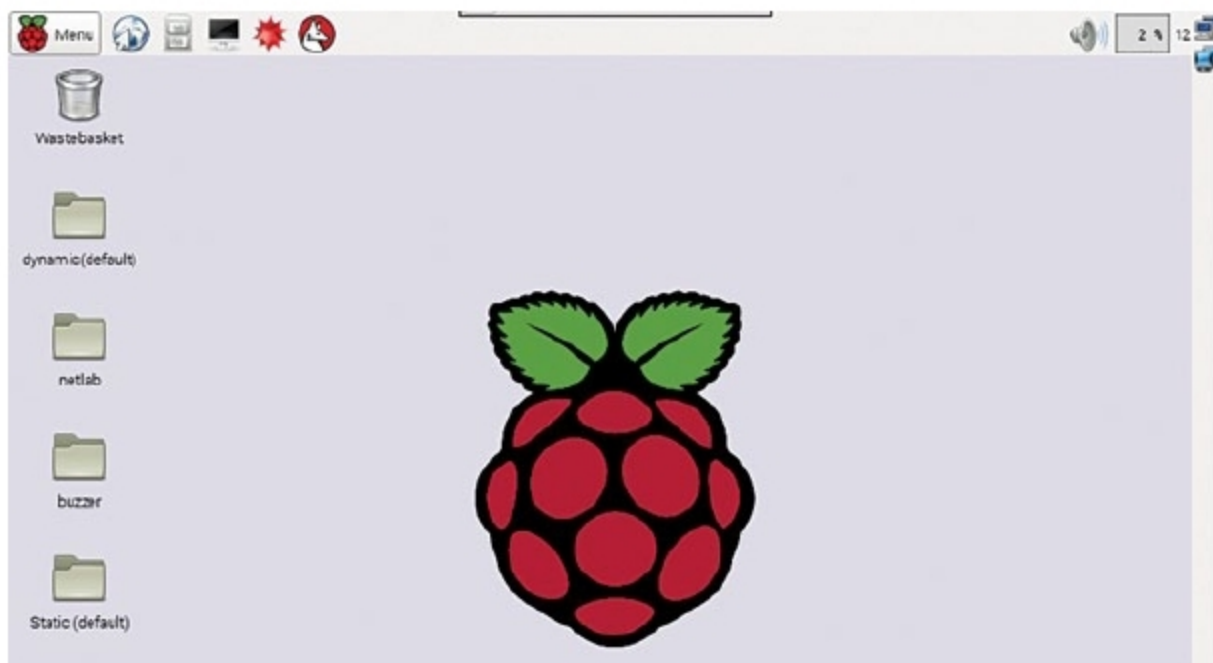


Fig. 6: Raspberry Pi desktop



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